Junpyo Jang

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RESEARCH INTERESTS

- **Materials & phenomena:** Spin textures in 2D van der Waals (vdW) magnets (stripe domains, skyrmion lattices) and their control by moiré patterns, strain, and ionic intercalation
- **Methods & devices:** Quantitative and in-situ electron microscopy with custom image analysis; integration of 2D magnetic materials into beyond-CMOS memory and logic

EDUCATION

Korea University, Seoul, Republic of Korea — Mar 2017 – Feb 2023
B.S. in Materials Science and Engineering
GPA: 3.95/4.5 (Major: 4.27/4.5; Junior/Senior: 4.29/4.5)
Leave for compulsory military service (Auxiliary Police, Seoul Metropolitan Police Agency) — *Oct 2018 – May 2020*

RESEARCH EXPERIENCE

Research Intern *(Full-time)* — Apr 2025 – Present
Quantum-material Spintronics Group (Advisor: Jun Woo Choi)
Center for Semiconductor Technology, Korea Institute of Science and Technology (KIST)

Moiré-modulated magnetic textures in Fe_3GaTe_2

- Study, using Lorentz TEM (LTEM), how the local moiré geometry, created by slight interlayer strain and twist within a single flake of the room-temperature vdW ferromagnet Fe_3GaTe_2, relates to stripe domains (zero-field cooled), skyrmion lattices (field cooled), and their response to field
- Prepare samples on TEM grids, measure MOKE, plan and take part in TEM and LTEM measurements, and analyze both the moiré structure and the magnetic domains
- Established a same-flake, two-region comparison protocol that separates moiré-related differences from flake-to-flake variation
- Built an LTEM analysis tool on PyLorentz: TIE phase and induction maps, stripe-domain period and orientation, and skyrmion-lattice statistics (density, spacing, hexagonal order, size)
- Built a moiré analysis tool that maps the local moiré period, direction, twist, and strain across a flake, and a browser-based moiré simulator that generates synthetic TEM images with known ground truth to validate it
- Poster at the 2026 MRS Fall Meeting; one manuscript in preparation

Li-ion intercalation in vdW magnets by solid-state ionic gating *(contributor)*

- Assist with FIB sample preparation and TEM measurements of cross-sectional and plan-view specimens of Fe_3GaTe_2 and CrSBr gated through lithium silicate, including in-situ gating experiments, to image Li-ion intercalation and its structural effects

Strain-tunable magnetism by polymer-based tensile strain *(contributor)*

- Fabricate polymer-supported samples and measure their response to applied and released uniform tensile strain by MOKE and Raman spectroscopy

vdW heterostructure devices

- Assembled asymmetric-stoichiometry all-vdW MTJ stacks ($Fe_{2.7}GeTe_2$/hBN/Fe_3GeTe_2) by glovebox exfoliation and dry transfer, including exfoliation of hBN tunnel barriers
- Fabricated a graphite/In_2Se_3/Fe_3GaTe_2/hBN ferroelectric gating stack for low-temperature MOKE measurements

PUBLICATIONS AND PRESENTATIONS

Journal manuscripts

[1] **J. Jang** et al., manuscript on moiré-modulated stripe domains in Fe_3GaTe_2, *in preparation*.

Conference presentations

[1] **J. Jang**, M. Jang, H. Suh, K. Kim, and J. W. Choi, "Moiré-Modulated Magnetic Textures in Fe_3GaTe_2 Revealed by Lorentz Transmission Electron Microscopy." Poster, 2026 MRS Fall Meeting & Exhibit (Symposium QT01), Boston, MA, Dec 2026.

INDUSTRY EXPERIENCE

Engineer *(Full-time)***, Logic Technology Development Team** Mar 2023 – Mar 2025
Semiconductor R&D Center, Samsung Electronics

1.4 nm GAA logic node (Samsung Foundry SF1.4): BEOL interconnect scheme development and integration

- Defined and validated via/metal stack and line-cut schemes to lower RC delay and defectivity, qualifying stable fab-out across critical development lots
- Achieved ~10% reduction in contact resistance (Rc) and resolved hard-mask etch drifts causing line-opens by engineering over-etch splits and stack-down margins, significantly reducing yield excursions
- Correlated inline metrology, electrical test data, and cross-sectional TEM to set pre/post-process checks that preempted excursions as new schemes entered development lots; designed BEOL mask TEGs and ran DRC/LVS (Calibre)

Backside power delivery network (BSPDN): backside BEOL route development and integration

- Set up a backside BEOL route for BSPDN, defining the process flow and specs to decouple power from signal routing and securing overlay/process windows with unit process teams; ran wafers through the route and verified it with electrical tests (ET)
- Achieved 100% landing yield of nano-TSV (nTSV) backside power vias to M1 by resolving the nTSV–M1 alignment challenge (Quarterly Team Performance Recognition)

Intern *(Full-time)***, Logic Technology Development Team, Samsung Electronics** Jul 2022 – Aug 2022

- 1.0 nm logic node: analyzed Ru-based BEOL interconnects and mapped end-to-end integration flows as a reference for sub-nm technology

TECHNICAL SKILLS

- **Electron microscopy:** Lorentz TEM and TEM/SAED (planning and taking part in measurements; data analysis), TIE phase reconstruction, SEM; FIB sample preparation and in-situ TEM (assisting)
- **Analysis & computation:** Python (image-analysis pipelines: FFT, lock-in/GPA phase mapping, Voronoi/ψ_6 statistics, image registration); browser-based simulation tools (HTML/JavaScript); MuMax3 (with collaborator); DigitalMicrograph (GMS 3); Origin; Spotfire; VESTA
- **2D materials & devices:** Mechanical exfoliation and dry transfer (glovebox), vdW heterostructure assembly, e-beam/photolithography, thermal evaporation, sputtering, ion milling, wire bonding, polymer-based strain application
- **Characterization:** MOKE, Raman spectroscopy, VSM, PPMS, AFM, probe station
- **Semiconductor process & CAD:** BEOL process integration, Calibre (DRC/LVS), KLayout, nanoCAD

HONORS AND AWARDS

Quarterly Team Performance Recognition, Logic Technology Development Team, Samsung Electronics Nov 2023
Semester High Honors (Academic Excellence), College of Engineering, Korea University 4 semesters, 2020–2022

SERVICE AND OUTREACH

Vice-President, Korea University MSE Undergraduate Society (KUMUS) Fall 2021 – Spring 2022

- Organized weekly student-led seminars on faculty lab research and coordinated student participation in industry events (SEDEX)

Volunteer Tutor (65 hours), Seomna Child Welfare Center Mar 2019 – Jan 2020

- One-on-one after-school math and science mentoring for underserved elementary and middle-school students